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Engineering Innovation Challenge 2023

Significant - Sign Language App for Social Inclusivity & Well-Being

Team U-17

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Project Abstract

In 2022, approximately 1% of people worldwide rely on sign languages as their primary means of communication. This poses a challenge when interacting with the remaining 99% of the population who do not understand sign language. Therefore, it is important to cultivate an inclusive society which welcomes and embraces individuals despite their disabilities. To tackle this concern, Team U-17 has come out with a software application, ***Significant*** (Sign-if-i-can't).

To demonstrate our solution and its feasibility, the ***Significant*** prototype has been developed. It encompasses several functionalities, including the translation of sign language into spoken language, Speech-to-Text, and Text-to-Speech. Sign languages used by the deaf community can be translated into spoken languages through ***Significant***, which empowers them to express themselves to the hearing community. ***Significant*** offers a two-way conversion between speech and text, which allows individuals to choose their preferred mode of input.

The prototype is developed with Python as its programming language, utilising various resources such as Mediapipe, OpenCV, Keras, Tkinter, Google Translate, Google Text-To-Speech, and ChatGPT to achieve its functions. However, the development of the prototype brought to light its limitations such as the need for a substantial dataset, and the implementation of the software in a mobile setting.

We aim to make sign languages become more accessible like spoken languages and reduce communication barriers and social isolation for the global sign language user community. Individuals are able to express themselves without language barriers by utilising technology in public speaking scenarios. Eventually, it promotes diversity and inclusivity in resilient cities.

Project Report

1. Background

Sign language serves as a visual form of communication employed by individuals who are deaf or have hearing difficulties to interact both within their community and with the hearing population. Recent reports from 2022 indicate that sign languages are relied upon by approximately 1% of the global population (72M) as their primary method of communication (National Geographic, 2022). Consequently, there exists a widespread lack of familiarity with sign language among the general public, leading to significant communication obstacles between the deaf and hearing communities. Adding to this challenge, different countries have developed their distinct sign languages, such as American Sign Language (ASL), British Sign Language (BSL), and Singapore Sign Language (SgSL).

Communication challenges frequently lead to emotional distress and social isolation. In one study involving hearing-impaired individuals, 41% of them believed that communication problems coupled with family stresses and overall prejudice could cause or contribute to suicidal depression, substance abuse or violent behaviour in some cases. (Purse, 2021) This highlights the crucial need to foster social inclusivity by bridging these communication gaps.

2. Purpose

Our mission is to cultivate an inclusive society where every individual, regardless of their disabilities, is welcomed and embraced. Only by ensuring that no one is left behind can we birth a resilient city, ready to confront all challenges.

3. Prototype Design and Concept

Introducing our prototype, the mobile app *Significant*. Designed to bridge communication between sign language users and non-users, it features a computer vision sign language translator coupled with ChatGPT's language processing. This dynamic combo seamlessly converts gestures to various spoken languages. For non-sign language users, easy text input or voice recording is translated into written text in any language.

It has the potential to significantly improve communication access and inclusion for hearing-impaired individuals in real-time without an interpreter, promoting greater understanding and collaboration between hearing and deaf communities.

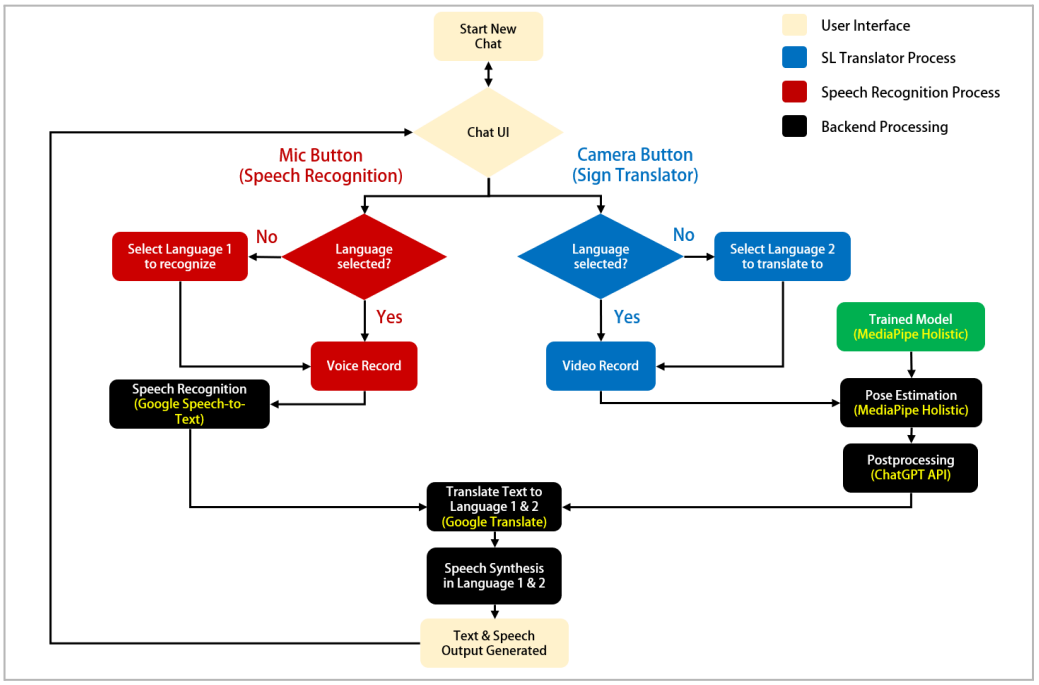


Figure 1. Architectural Framework

3.1 App User Interface (UI)

a. Launch the ***Significant*** app and initiate a new chat

b. Enter the chat and select Languages 1 and 2. Both communicators should be comfortable with at least one of the languages selected.

c. Select "Cam" for sign language translation or opt for "Voice" for speech recognition.

d. Once complete, detected sentences and their translations are displayed in a chat log.

e. Tap a chat bubble to hear the message read aloud via text-to-speech.

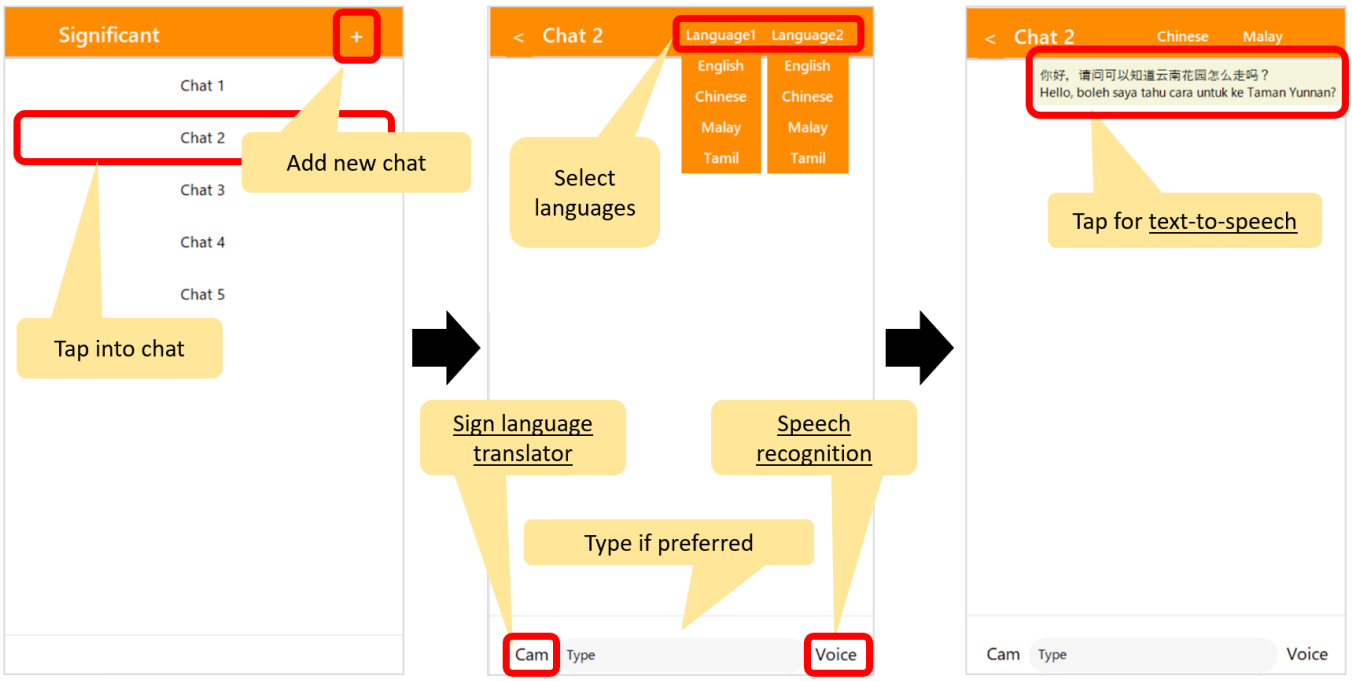


Figure 2. *Significant* User Interface

3.2 Sign Language to Spoken Language

When "Cam" is selected, ***Significant*** initiates a sign language translator and emerges a camera view for gesture detection.

Significant facilitates the conversion of American Sign Language (ASL) into text using motion tracking and action detection techniques. At the time of proposal submission, four phrases ("Hello", "How are you?", "I am fine", and "Thank you") have been incorporated with more to be added. Figure 2. showcases a successful recognition of the sign for "Hello".

Given the distinctive grammar and sentence structure of ASL compared to traditional spoken languages, the detected signs undergo additional refinement through ChatGPT's processing capabilities and are outputted as sentences complete with accurate punctuation and grammar, accompanied by their corresponding translations (Figure 3).

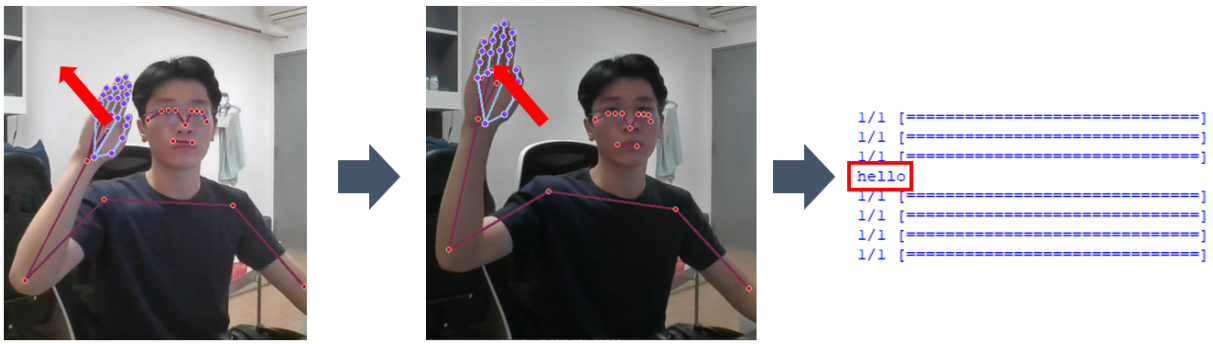


Figure 3. Sign detection of "Hello"

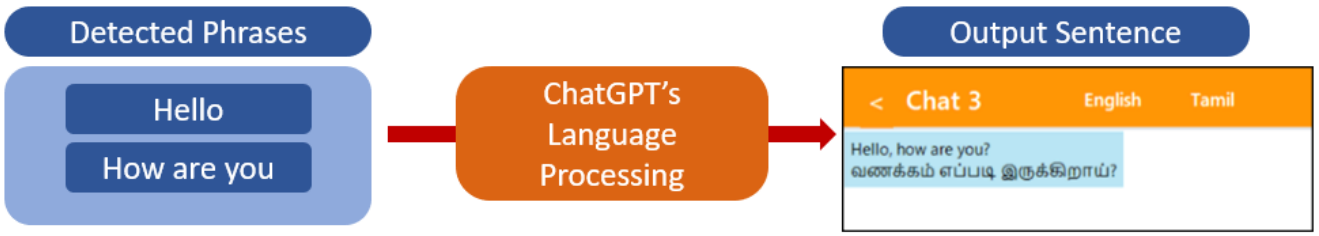


Figure 4. Sentence output after sign language translation & ChatGPT processing

3.3 Speech-to-Text

When "Voice" is selected, *Significant* initiates speech-to-text and voice recording will begin.

Significant is able to convert spoken language into text, facilitating communication with hearing-impaired individuals. Alternatively, the non-sign language user has the option to type their message instead.

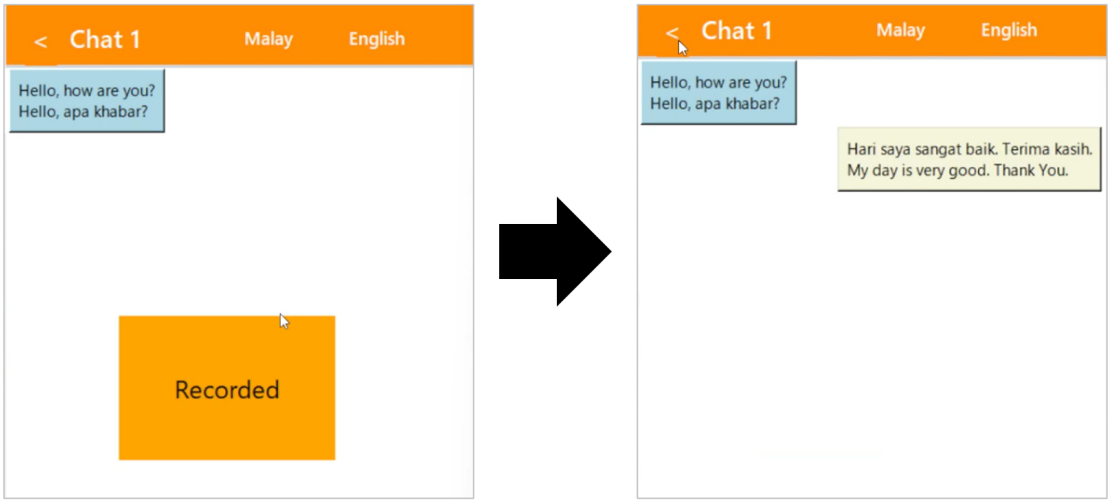


Figure 5. Sentence output after Speech-to-Text

3.4 Text-to-Speech

The text-to-speech function can be activated by tapping on any chat bubbles within the chat log, which can be crucial for those who have difficulties reading the text.

3.5 Chat History

The resulting text, whether converted or typed, is then translated and displayed to the other person in a chat log, enhancing communication accessibility. Similar to typical messaging apps, chat logs can be saved and accessed independently for each session, providing a comprehensive record of communication history for future reference.

4. Material

Significant is developed using Python as the primary programming language.

4.1 App UI

The user interface design of ***Significant*** is crafted using the Python Tkinter library, with the goal of replicating the familiar design elements found in typical messaging apps. All of the *Significant* features can be easily accessed and executed within this UI.

4.2 Sign Language to Spoken Language

Significant leverages the power of Google's Mediapipe and OpenCV for sign language detection. Additionally, the sign language detection model is trained using Keras, an open-source neural network library designed for Python.

4.3 Speech-to-Text

In order to transcribe the users' speech, Google Speech-to-Text API is used, as it supports more languages and has higher accuracy than other Speech-to-Text APIs like Sphinx and AssemblyAI.

4.4 Text-to-Speech

Significant uses Google Text-to-Speech API to generate audio from the text input and when the chat box is clicked.

4.5 Multiple Language Translation

To support a diverse array of languages, the construction of language translation relies on the utilisation of Google Translate API, one of the world's largest machine translation APIs.

4.6 Grammar and Punctuation Check

In order to strengthen the robustness of our sign language to spoken language transcriber, ChatGPT API is used to eliminate irrelevant words detected, ensuring grammatical accuracy and fluidity. Furthermore, the punctuation of the sentences is also added after speech-to-text and pose-to-text translations, ensuring that the transcriptions are easily comprehensible.

5. Limitations

a. A substantial dataset of sign language gestures is required to accurately train the model, which presents a challenge to build a comprehensive dictionary of signs in a short timeframe.
b. The prototype is developed on a computer and operates using Python. Additional work is required to adapt the program for mobile devices, transforming it into a functional mobile app.

6. Risk and Safety

a. Privacy Concern - To protect privacy, the process involves obtaining explicit user consent, collecting only essential data, encrypting captured information, and adhering strictly to privacy regulations.
b. Accuracy and Reliability - To prevent information loss caused by language entropy, a language processing model (ChatGPT) is utilised to convert the grammar of sign language into spoken language.

7. Conclusion

Our ultimate goal is to ensure that sign language becomes as accessible as other spoken languages through *Significant*, similar to how Google Translate works for spoken languages. Hence, communication barriers and social isolation can be significantly reduced or even eliminated for the deaf community worldwide. Moreover, the foundational technology of our sign language translation has a vast potential for expansion to various contexts. It can be harnessed for scenarios like public speaking and online meetings, empowering deaf individuals to express themselves without any language barriers. This will be a transformative step for humanity in fostering resilient cities that embrace diversity and inclusivity.

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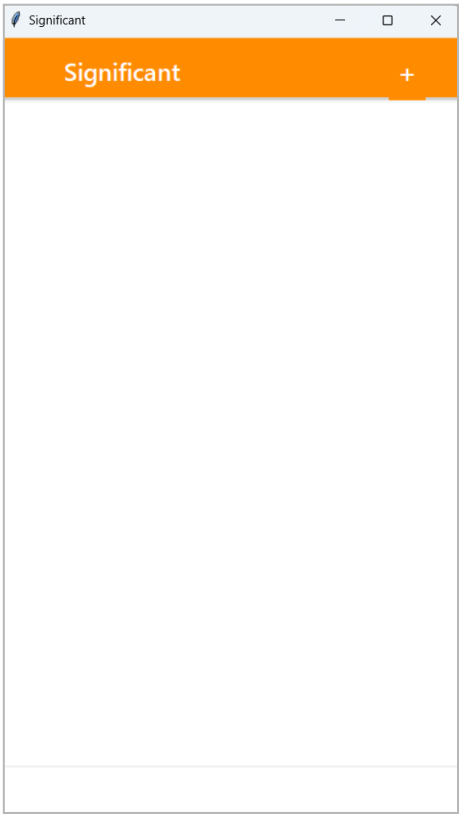
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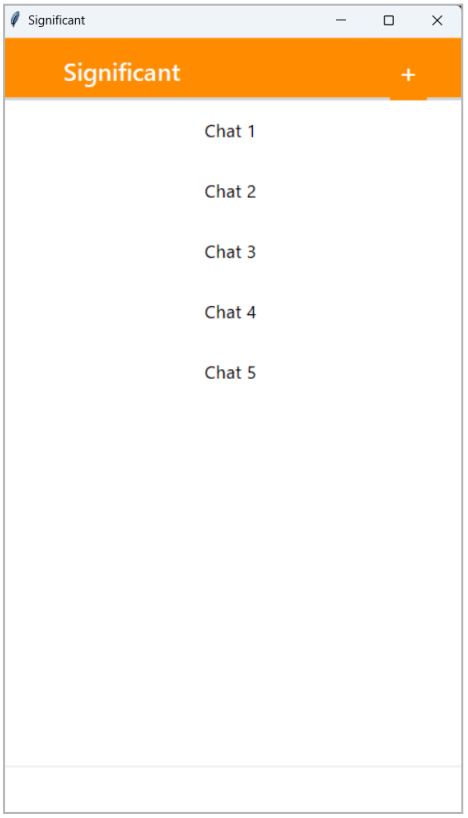
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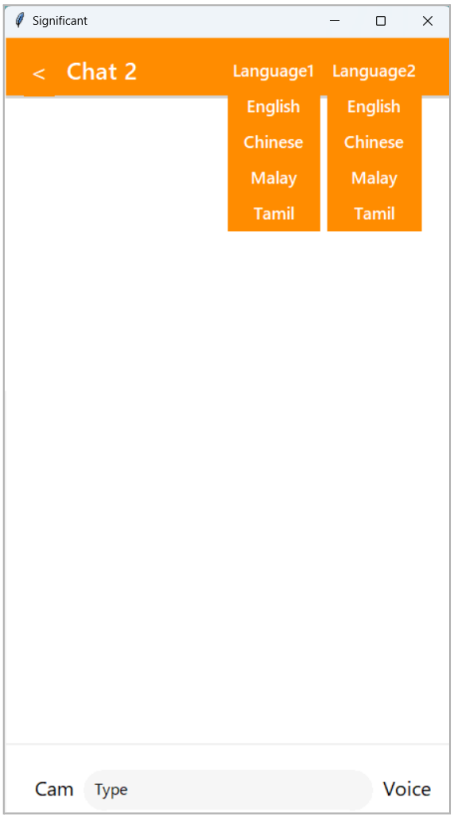
Appendix: Screenshots of the *Significant* app

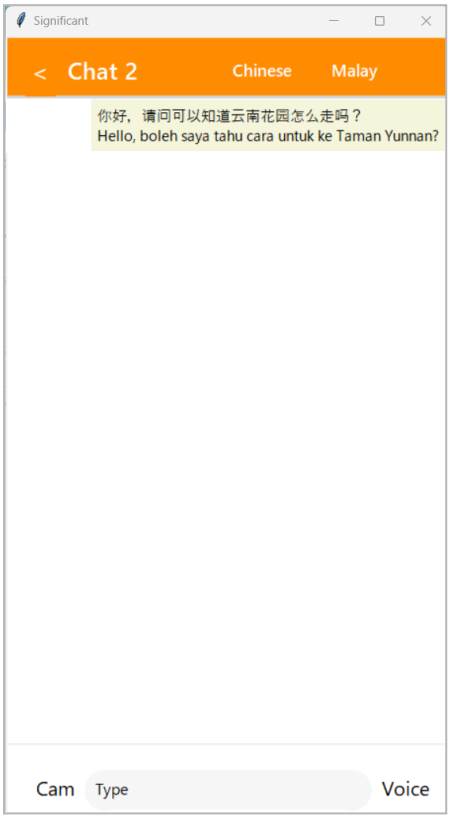
I. Main Interface



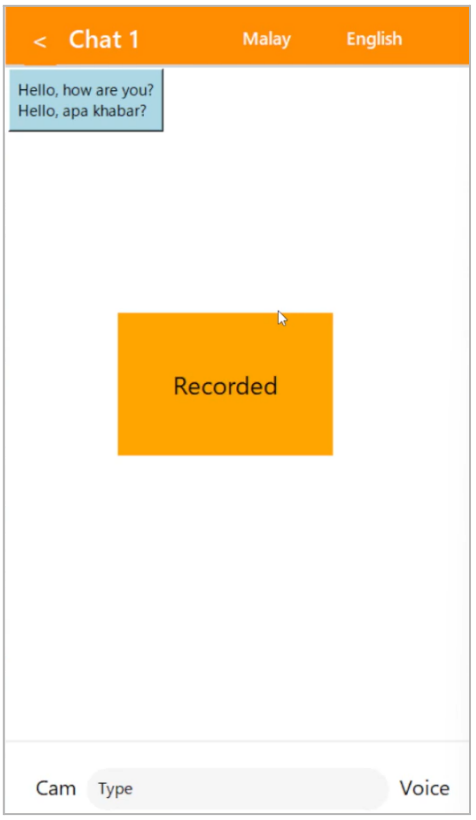


II. Chat Logs



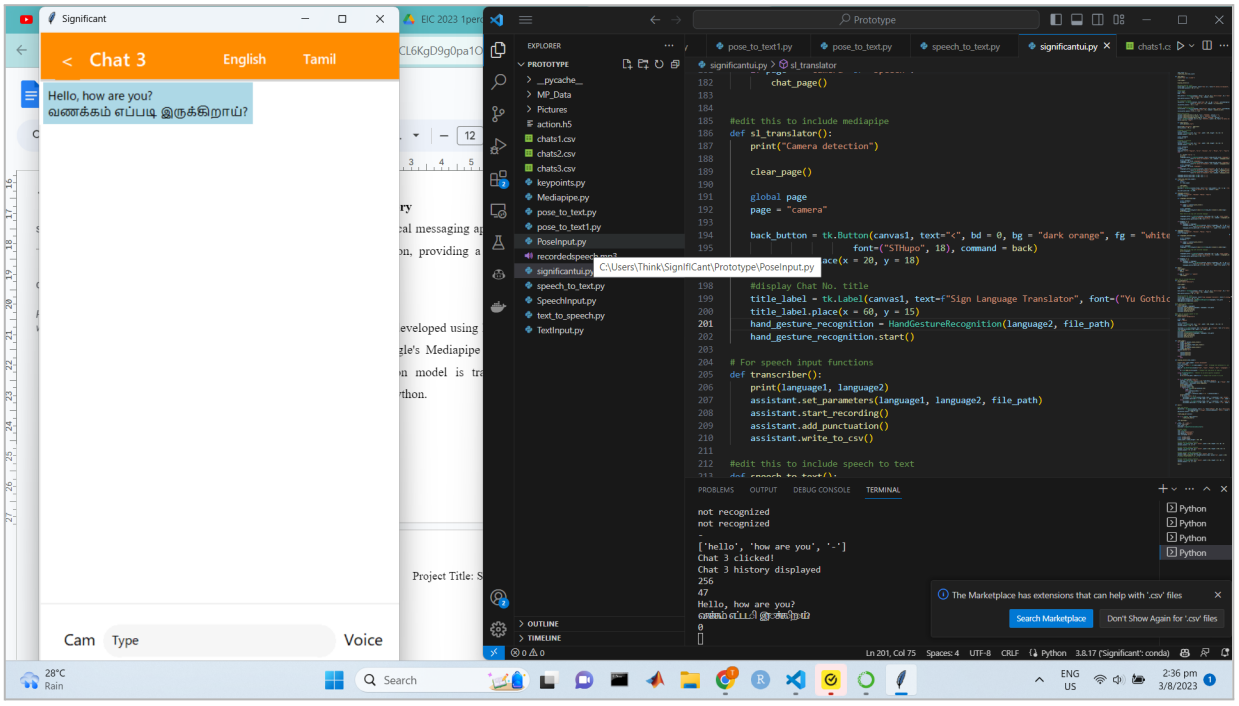


III. Speech-to-Text





IV. Sign Language to Spoken Language - with Backend



V. Speech-to-Text - with Backend

